Case G — Mr. Arthur Chen | 69M | Idiopathic Pulmonary Fibrosis (IPF)

Retired stonemason. Dry non-productive cough, | progressive breathlessness now limiting dressing. Digital clubbing. Fine

Target: 70%+ | Cardiorespiratory Physiotherapy

■ PROBLEM LIST

- Reduced lung compliance (stiff fibrotic lungs)
- Impaired gas exchange — reduced DLCO
- Exercise-induced hypoxaemia / exertional desaturation
- Reduced respiratory muscle endurance
- Progressive deconditioning and reduced aerobic capacity
- Digital clubbing (marker of chronic hypoxia)
- Breathlessness limiting dressing and personal care
- Reduced walking tolerance and community mobility
- Difficulty with stairs and inclines
- Reduced participation in leisure / social activities
- Anxiety related to breathlessness (breathlessness-fear cycle)
- Reduced quality of life

■ RED FLAGS

None identified

■ PATHOPHYSIOLOGY

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■ DIFFERENTIAL DIAGNOSES

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■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

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■ TREATMENT PLAN

■ Airway Clearance

- Not indicated as a primary technique in IPF.
- Mr. Chen has a dry, non-productive cough — IPF does not produce excess
- mucus, so ACBT or PEP would not be appropriate and may exacerbate cough
- distress. If cough is particularly distressing, refer to respiratory
- physician for pharmacological cough suppression (e.g., low-dose

■ Breathing Exercise: Pursed-Lip Breathing (PLB) + Paced Breathing

- Breathe in through the nose for
- 2 counts. Breathe out slowly through pursed lips (as if cooling hot
- soup) for 4 counts. Aim to reduce resting RR to 12–16 breaths/min. Use
- during all activities that provoke breathlessness.
- Practise supine with knees supported — eliminates postural demand.

■ Exercise & Mobility: Graded Walking Programme

- Starting level:
- Begin at a pace
- maintaining Borg dyspnoea ≤3/10 and SpO2 ≥90% throughout. Initial target
- may be as short as 5–10 minutes at slow pace. Use supplemental O2
- during exercise if exertional desaturation below 88% was confirmed on

■ Energy Conservation Strategies

- The Four Ps — Plan, Prioritise, Pace, Position:
- Break tasks into stages. Prioritise essential activities and delegate
- or simplify others. Plan rest periods before, not just after, demanding
- Positioning for breathlessness relief:
- Forward-lean sitting (elbows on knees or table) — fixes the shoulder